LESSON 09 · CHEAT SHEET

pandas I: DataFrames

IN AND OUT

```
import pandas as pd
pd.set_option("display.width", 120)
df = pd.read_csv("data/file.csv")
df = pd.read_excel(path, sheet_name="Quests")
df.to_csv("my_work/out.csv", index=False)
df.to_excel("my_work/out.xlsx", index=False)
```

LOOK AT IT

```
df.shape          (400, 12)  # rows, columns
df.columns        # column names
df.head()  df.head(3)  df.tail(3)
df.info()         # types + missing values
df.describe()     # count mean std min 25% ...
len(df)           400
```

COLUMNS

```
df["reward"]                  # one column (Series)
df[["quest_id", "monster"]]    # several (list!)
df["reward"].mean()            # .median() .max()
df["reward"].sum()             # .min() .std()
df["monster"].unique()         # distinct values
df["monster"].nunique()        # how many distinct
df["new"] = df["a"] / df["b"]  # add a column
df["job"] = df["monster"] + " in " + df["region"]
df = df.drop(columns=["tmp"])
df = df.rename(columns={"old": "new"})
```

FILTER ROWS (AUTOFILTER)

```
df[df["reward"] > 2000]
df[(df["region"] == "Greyfen") & (df["days"] < 5)]
df[(df["monster"] == "Wyrm") | (df["days"] > 9)]
df[df["region"].isin(["Greyfen", "Saltmarsh"])]
df[df["title"].str.contains("old mill")]
df[df["outcome"] != "Success"]
df[~mask]                      # not
```

SORT

```
df.sort_values("per_day")          # low to high
df.sort_values("per_day", ascending=False)
df.sort_values(["region", "reward"],
               ascending=[True, False])
df.nlargest(5, "per_day")   df.nsmallest(3, "days")
```

ROWS BY LABEL / POSITION

```
df.loc[5]            # row with index label 5
df.iloc[0]           # first row by position
df.iloc[-1]          # last row
df.loc[5, "reward"]              # one cell
df.loc[mask, "paid"] = 0         # set via mask
df = df.reset_index(drop=True)   # renumber 0..n
```

DATES

```
df["posted"] = pd.to_datetime(df["posted"])
df["posted"].dt.year  .dt.month  .dt.day_name()
df[df["posted"] >= "2025-06-15"]  # compare
(df["posted"].max() - df["posted"].min()).days
```

EXCEL → PANDAS

a sheet	DataFrame `df`
a column	Series `df["col"]`
AutoFilter	`df[mask]`
Sort A→Z	`.sort_values("col")`
helper column	`df["new"] = ...`
COUNTA / COUNTIF	`len(df)` / `mask.sum()`
Save As CSV	`.to_csv(..., index=False)`

GOTCHAS

- Column names are case-sensitive: `KeyError: 'Region'` means check the spelling
- Two columns need a list: `df[["a", "b"]]`, double brackets
- `sort_values` returns a copy; assign it or chain it
- Forgot `index=False`? Hello, "Unnamed: 0"
- Masks: `&` `|` `~` with brackets, never `and` `or` `not`